

JOB PORTAL MANAGEMENT SYSTEM

School of Computing

**Bachelor of Technology
in
Computer Science & Engineering**

By

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May, 2026

BONAFIDE CERTIFICATE

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DECLARATION

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ABSTRACT

The Job Portal Management System is a full stack web-based application designed to streamline the recruitment process by connecting job seekers and employers on a unified digital platform[3]. Traditional job searching and hiring methods often involve manual processes, delayed communication, and lack of transparency, which reduce efficiency for both candidates and recruiters. The proposed system allows users to create profiles, upload resumes, search and apply for jobs, and track application status through an intuitive and user-friendly interface. Employers can post job openings, manage applications, and shortlist candidates efficiently. User experience and a robust backend system for secure data processing and storage. It incorporates advanced features such as real-time email verification using SMTP for secure user authentication and an integrated chatbot to assist users with queries, navigation, and job-related guidance. The system ensures accurate data management by maintaining structured records of users, job postings, and applications in a relational database.

Keywords: Job Portal, Spring Boot, REST API, Spring MVC, JPA, MySQL, Spring Security, Email Verification, Chatbot, Web Application, CRUD Operations.

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LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
CSS	Cascading Style Sheets
HTML	HyperText Markup Language
HTTPS	HyperText Transfer Protocol Secure
JWT	JSON Web Token
OTP	One-Time Password
SMTP	Simple Mail Transfer Protocol
SQL	Structured Query Language
UI	User Interface
MVC	Model View Controller
CRUD	Create, Read, Update, Delete
DBMS	Database Management System

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Chapter 1

INTRODUCTION

1.1 Introduction

The Job Portal Management System is a web-based application developed to simplify and modernize the recruitment process by connecting job seekers and employers on a single digital platform. Traditional job searching and hiring processes often rely on manual methods, which lead to inefficiencies, delayed communication, and lack of transparency. This system provides a centralized solution where job seekers can create profiles, upload resumes, browse job listings, and apply for jobs, while employers can post job openings, review applications, and manage candidates efficiently.

The application integrates modern web technologies to ensure a responsive and user-friendly experience. It also incorporates advanced features such as an intelligent chatbot to assist users with queries and navigation, and secure email verification using SMTP-based OTP au-

thentication to ensure user authenticity. By automating recruitment workflows and improving communication, the system enhances overall efficiency and user satisfaction.

1.2 Aim of the Project

The aim of this project is to design and develop a secure, efficient, and user-friendly Job Portal Management System that automates the recruitment process. The system focuses on reducing manual effort, improving job accessibility, and enabling seamless interaction between job seekers and employers. It also aims to provide secure authentication through email verification, intelligent assistance through chatbot integration, and real-time updates for better user engagement and decision-making.

1.3 Scope of the Project

The scope of the Job Portal Management System includes user registration, profile management, resume upload, job posting, job search and filtering, and application tracking. Job seekers can explore job opportunities, apply online, and monitor their application status, while employers can create, update, and manage job listings and evaluate candidates.

The system also includes additional features such as chatbot support for user guidance and email verification for secure account management. The application is primarily web-based and designed for scalability. Future enhancements may include mobile application support, AI-based job recommendations, and advanced analytics for recruitment insights.

1.4 Framework

The system is developed using the Spring Boot framework, which provides a robust backend environment for building scalable and secure web applications. Spring MVC is used to implement the Model-View-Controller architecture, ensuring proper separation of concerns. Spring Data JPA is utilized for efficient database interaction and ORM-based data handling.

The frontend is developed using HTML, CSS, JavaScript, and Thymeleaf to create dynamic and responsive user interfaces. MySQL is used as the relational database for storing user, job, and application data. Spring Security is implemented to handle authentication and authorization, along with password encryption for enhanced security. Additionally, SMTP-based email services are integrated for OTP verification, and a chatbot module is incorporated to provide interactive user

support. This combination of technologies ensures that the system is scalable, secure, and maintainable.

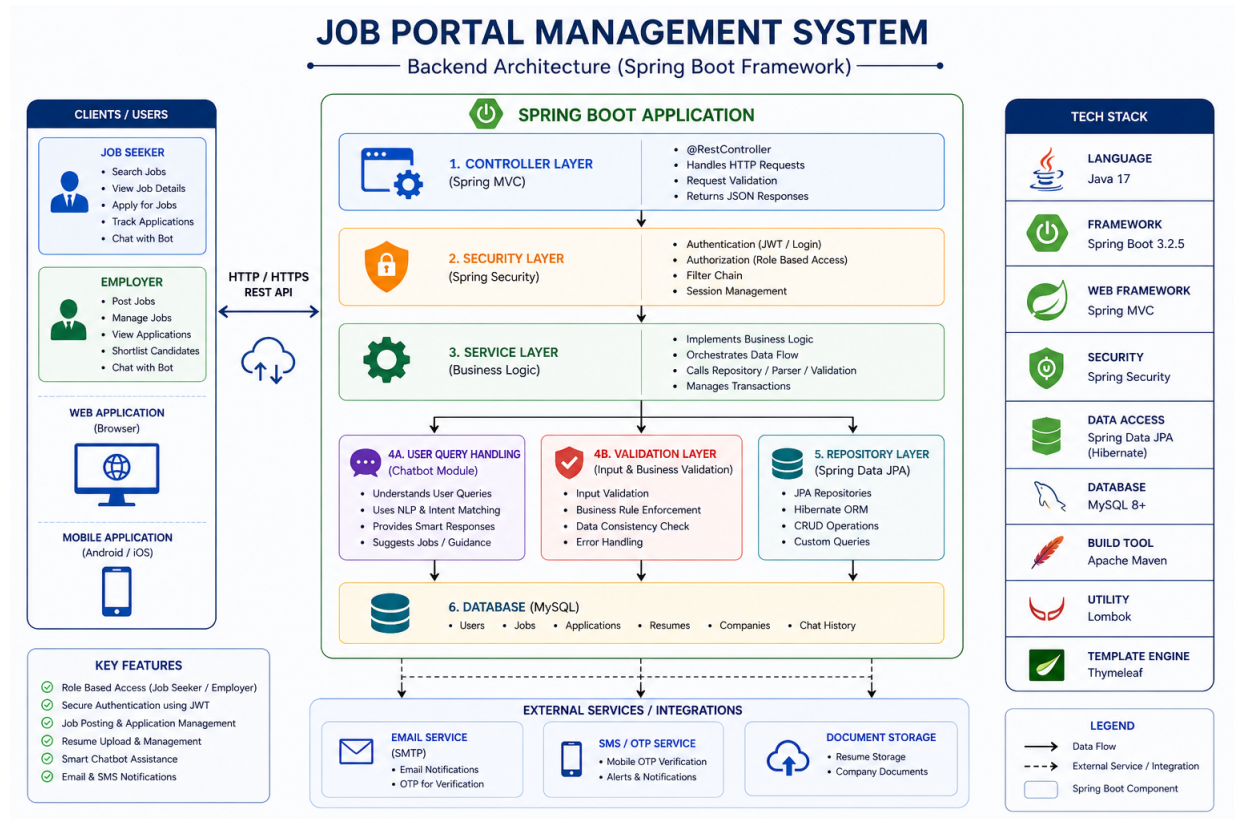


Figure 1.1: Framework and Architecture

[Figure 1.1] presents the overall framework and architecture of the Job Portal Management System built using Spring Boot. It shows how different layers such as controller, security, service, validation, and repository work together to process user requests. The system also integrates external services like email verification, chatbot assistance, and resume storage. All data is managed in a MySQL database, ensuring secure and efficient operation of the application.

Chapter 2

SURVEY OF SIMILAR APPLICATION IN MARKET

2.1 Global Job Portals

Leading platforms such as LinkedIn and Indeed dominate the global recruitment ecosystem by providing extensive job listings, professional networking features, and simplified application processes[1]. These platforms enable large-scale interaction between recruiters and job seekers through user-friendly interfaces and advanced filtering options. However, most of these platforms focus primarily on job listings and networking, with limited personalized assistance for users[2].

2.2 Company Insight Platforms

Platforms like Glassdoor provide valuable insights into company culture, salary structures, and interview experiences. These features

help candidates make informed career decisions beyond basic job applications[3]. While such platforms enhance transparency, they often lack integrated recruitment functionalities such as direct application tracking and interactive support systems.

2.3 Indian Job Portals

In the Indian context, job portals such as Naukri.com and Internshala are widely used by both fresh graduates and experienced professionals[2]. These platforms offer targeted job listings, internship opportunities, and skill-based recommendations. Although they are effective in connecting employers and candidates, they generally lack advanced features like real-time assistance and enhanced security mechanisms such as OTP-based verification[4].

2.4 AI-Driven Recruitment Platforms

Modern recruitment platforms like Monster and ZipRecruiter incorporate advanced technologies such as resume parsing and AI-driven job recommendations[3]. These systems improve hiring efficiency by automating candidate matching and filtering processes. However, they often require premium access for advanced features and may not provide integrated support systems like chatbots for continuous user in-

teraction[5].

2.5 Identified Research Gap

Despite the availability of various job portals and recruitment platforms, several limitations still exist. Most existing systems focus on individual functionalities such as job listing, networking, or analytics rather than providing a unified solution[4]. There is a lack of platforms that combine secure authentication mechanisms, real-time user assistance, and interactive communication features.

The proposed Job Portal Management System addresses these gaps by integrating email verification using SMTP-based OTP authentication for enhanced security and an intelligent chatbot to assist users with queries and navigation. Additionally, it provides a centralized platform with application tracking, job filtering, and role-based access control, ensuring a more efficient and user-friendly recruitment process[5].

2.6 Comparison of Existing Systems

To better understand the competitive landscape, a comparison of existing job portals with the proposed system is performed based on key features such as security, user assistance, and system functionality. The proposed system stands out by offering integrated chatbot support,

secure email verification, and a unified platform for both job seekers and employers, which are not commonly available together in existing solutions[4].

Table 2.1: Feature Comparison with Market Leaders

Feature	LinkedIn	Naukri.com	Internshala	Proposed System
Job Application	High	High	High	High
AI Interview/Question Bot	No	No	No	Yes
Modern Responsive UI Design	Yes	Basic	Basic	Yes
Integrated Support Chatbot	Limited	No	No	Yes
SMTP Email Verification	Yes	Yes	Yes	Yes
Application Tracking System	Yes	Yes	Yes	Yes
Real-time Notifications	Limited	Basic	Basic	Yes
Security (OTP + Authentication)	Partial	Partial	Partial	Strong
Real-time Analytics	Paid	Basic	Basic	Built-in

[Table 2.1] presents a comparison of existing job portal platforms with the proposed system based on key features. It highlights that while existing platforms provide core job application functionalities, they often lack integrated intelligent assistance and enhanced security mechanisms[7]. The proposed system stands out by incorporating chatbot-based user support, SMTP-based email verification for secure authentication, real-time notifications, and built-in analytics. These features collectively improve user experience, system security, and overall recruitment efficiency[8].

Chapter 3

FRAMEWORK DESCRIPTION

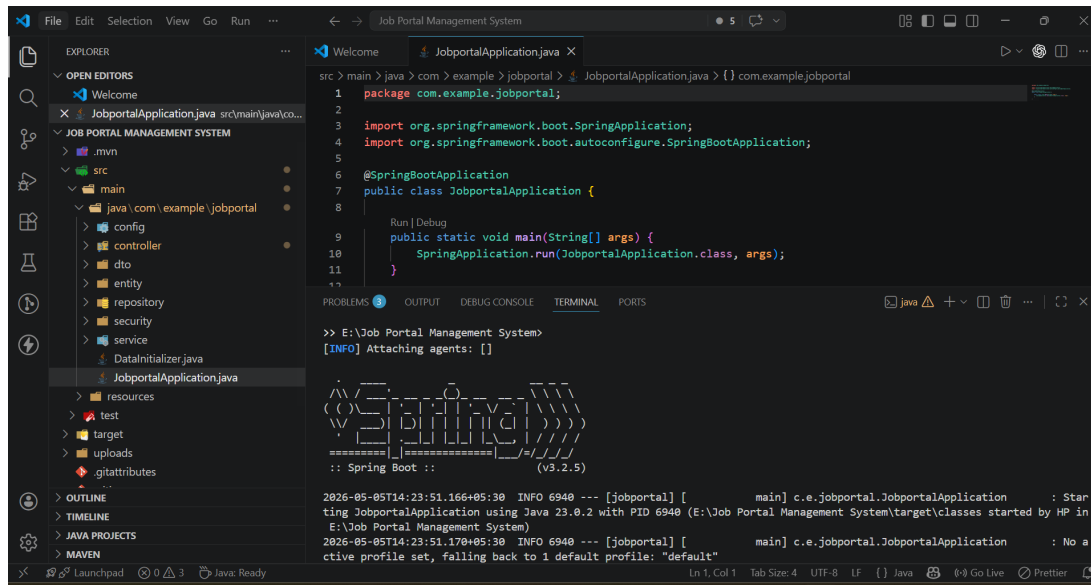
3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend of the Job Portal Management System is developed using HTML5, CSS3, JavaScript, and Thymeleaf. The development environment includes tools such as Visual Studio Code for coding and browser-based testing. Thymeleaf is used as a server-side template engine that integrates with Spring Boot to render dynamic content.

Basic styling is implemented using CSS to provide a clean and responsive user interface. JavaScript is used to enhance interactivity, such as dynamic job filtering and form validations. The system ensures a smooth and user-friendly experience for both job seekers and employers.

3.1.2 Structure of the Project



This image represents the development environment of the Job Portal Management System using Spring Boot in Visual Studio Code. It shows the project structure organized into layers such as controller, service, repository, and security for modular development. The terminal output confirms successful application startup, indicating that the backend server is running properly.

3.1.3 Execution Procedure

The frontend is executed through the Spring Boot application. When the server runs, Thymeleaf templates are rendered dynamically based on user requests handled by controllers. Users can access the application through a web browser using a local server URL (e.g., <http://localhost:8080>).

Navigation between pages is handled through controller mappings,

and user inputs are processed via forms integrated with backend services. Features such as job search, application submission, and chatbot interaction are handled dynamically.

3.2 Backend Implementation

3.2.1 Environment Setup

The backend is developed using Java and the Spring Boot framework. The development environment includes JDK 17 or higher, Maven for dependency management, and IDEs such as IntelliJ IDEA or Visual Studio Code.

MySQL is used as the database and is configured through the `application` file. Dependencies such as Spring Web, Spring Data JPA, Spring Security, and Validation are included to support backend functionalities. Additional configurations are included for SMTP-based email verification and chatbot integration.

Thymeleaf templates are rendered dynamically based on user requests handled by controllers. Users can access the application through a web browser using a local server URL (e.g., `http://localhost:8080`). Navigation between pages is handled through controller mappings, and user inputs are processed via forms integrated with backend services.

3.2.2 Structure of the Project

ID	ACTIVE	CATEGORY	CREATED_AT	DEADLINE	DESCRIPTION	EXPERIENCE_LEVEL	JOB_TYPE	LOCATION	MAX_SALARY	MIN_SALARY	SALARY	TITLE
1	TRUE	Engineering	2026-05-05 14:32:06.574942	2026-07-12	Design and maintain high-performance Spring Boot services. Improve API reliability, and mentor junior engineers.	3-5 Years	Full-time	Bangalore, India	2800000	1800000	₹18L - ₹23L	Senior -
2	TRUE	DevOps	2026-05-05 14:32:06.575947	2026-06-14	Manage CI/CD pipelines, AWS infrastructure, Kubernetes clusters, Terraform modules, and production observability.	2-4 Years	Full-time	Hyderabad, India	2400000	1400000	₹14L - ₹24L	DevOps
3	TRUE	Management	2026-05-05 14:32:06.581174	2026-06-13	Own product discovery, prioritize roadmap work, coordinate engineering delivery, and turn customer insights into measurable releases.	4+ Years	Remote	Remote	3500000	2200000	₹22L - ₹35L	Product
4	TRUE	Engineering	2026-05-05 14:32:06.581174	2026-06-30	Build Selenium and API automation suites, improve regression coverage, and partner with developers on release quality.	1-3 Years	Full-time	Chennai, India	1400000	800000	₹8L - ₹14L	QA/Auto
5	TRUE	Engineering	2026-05-05 14:32:06.584956	2026-06-19	Assist with Java APIs, SQL queries, unit tests, and internal tooling while learning production engineering practices.	0-1 Years	Internship	Bangalore, India	500000	300000	₹3L - ₹5L	Backen
6	TRUE	Engineering	2026-05-05 14:32:06.584956	2026-06-19	Build responsive interfaces with React, TypeScript, reusable components, and REST API integrations.	1-3 Years	Full-time	Mumbai, India	1600000	900000	₹9L - ₹16L	Fronten
7	TRUE	Data & AI	2026-05-05 14:32:06.584956	2026-06-11	Analyze large datasets, build predictive models with Python, and deploy ML experiments for business teams.	2-5 Years	Full-time	Pune, India	2600000	1500000	₹15L - ₹26L	Data Sc
8	TRUE	Design	2026-05-05 14:32:06.586024	2026-07-17	Create Figma prototypes, run usability tests, and collaborate with product managers on user-centered workflows.	1-2 Years	Remote	Remote	1300000	700000	₹7L - ₹13L	UX/UI D
9	TRUE	Marketing	2026-05-05 14:32:06.586024	2026-07-17	Plan acquisition campaigns, track funnel metrics, write landing page copy, and optimize paid channels.	1-3 Years	Full-time	Delhi, India	1100000	600000	₹6L - ₹11L	Growth
10	TRUE	Sales	2026-05-05 14:32:06.586024	2026-07-17	Qualify inbound leads, run product demos, maintain CRM hygiene, and help convert mid-market prospects.	0-2 Years	Full-time	Gurgaon, India	900000	500000	₹5L - ₹9L	Inside S
11	TRUE	DevOps	2026-05-05 14:32:06.588389	2026-07-03	Design cloud migrations, advise customers on secure architectures, and create scalable AWS and Azure reference patterns.	5+ Years	Full-time	Noida, India	4200000	2600000	₹26L - ₹42L	Cloud E
12	TRUE	DevOps	2026-05-05 14:32:06.590862	2026-07-18	Improve service reliability, tune alerting, automate incident response, and reduce toil across distributed systems.	3-5 Years	Remote	Remote	3000000	1800000	₹18L - ₹30L	Site Re
13	TRUE	Engineering	2026-05-05 14:32:06.590862	2026-07-07	Monitor security events, investigate incidents, manage vulnerability reports, and support compliance readiness.	2-4 Years	Full-time	Bangalore, India	1900000	1100000	₹11L - ₹19L	Secur
14	TRUE	Engineering	2026-05-05 14:32:06.592007	2026-06-14	Troubleshoot customer issues, document solutions, and collaborate with engineering on product improvements.	0-2 Years	Full-time	Kochi, India	800000	450000	₹4L - ₹8L	Technic
15	TRUE	HR	2026-05-05 14:32:06.592007	2026-06-22	Coordinate onboarding, maintain employee records, support policy rollouts, and improve hiring operations.	2-4 Years	Full-time	Pune, India	950000	550000	₹5L - ₹9L	HR/Op
16	TRUE	Finance	2026-05-05 14:32:06.594993	2026-06-24	Build forecasts, analyze monthly performance, prepare dashboards, and support business planning reviews.	1-3 Years	Full-time	Mumbai, India	1500000	800000	₹8L - ₹15L	Financ
17	TRUE	Management	2026-05-05 14:32:06.594993	2026-06-26	Map business requirements, write user stories, analyze data flows, and support Agile delivery teams.	2-4 Years	Full-time	Bangalore, India	1800000	1000000	₹10L - ₹18L	Busines
18	TRUE	Engineering	2026-05-05 14:32:06.594993	2026-06-25	Develop Android and iOS features with React Native, integrate APIs, and improve app performance.	2-5 Years	Full-time	Ahmedabad, India	2200000	1200000	₹12L - ₹22L	Mobile
19	TRUE	Marketing	2026-05-05 14:32:06.596996	2026-06-04	Own editorial calendars, publish SEO content, measure campaign performance, and refine brand messaging.	3-5 Years	Remote	Remote	2000000	1200000	₹12L - ₹20L	Content
20	TRUE	HR	2026-05-05 14:32:06.596996	2026-06-09	Schedule interviews, maintain candidate communication, update ATS records, and support hiring managers.	0-2 Years	Contract	Jaipur, India	650000	350000	₹3L - ₹6L	Recrui

This image represents the database view of the Job Portal Management System displaying records from the JOBS table. It shows various job details such as category, description, experience level, location, and salary information stored in the system. The interface confirms successful data storage and retrieval using the database, supporting efficient job management functionality.

3.2.3 Execution Procedure

The backend application is executed using Maven commands such as `mvn spring-boot:run`. Once the server starts, it listens for HTTP requests on a specified port (default: 8080).

The controllers process incoming requests, interact with the service and repository layers, and return responses to the frontend. Database

operations are performed using JPA, and data is stored and retrieved from MySQL.

Security mechanisms ensure that only authenticated users can access protected resources. Email verification is implemented using SMTP-based OTP validation, and the chatbot module assists users by providing automated responses to queries.

3.3 Database Implementation

3.3.1 Database Configuration

The Job Portal Management System uses MySQL as the relational database for storing and managing application data. The database is configured in the `application.properties` file of the Spring Boot project, where parameters such as database URL, username, password, and driver class are specified.

Spring Data JPA is used to interact with the database, enabling automatic table creation and simplified CRUD operations. Hibernate is used as the ORM (Object Relational Mapping) tool, which maps Java entity classes to database tables. This configuration ensures efficient data handling, scalability, and secure integration with the backend.

3.3.2 Schema Structure

The database schema is designed to support core functionalities such as user management, job posting, and application tracking. The main entities in the schema are User, Job, Application, Resume, and OTP.

The User entity stores information about job seekers and employers, including login credentials and roles. The Job entity contains details such as job title, description, required skills, salary, and location. The Application entity maintains the relationship between users and jobs.

The Resume entity stores information about uploaded resumes, while the OTP entity is used for email verification and password recovery. The schema follows relational database principles with proper primary and foreign key constraints to maintain data integrity.

3.3.3 Tables Created

The database includes the following main tables:

- **User Table:** Stores user details such as user ID, name, email, password, and role (job seeker/employer).
- **Job Table:** Contains job-related information including job ID, title, description, skills, salary, and location.
- **Application Table:** Maintains records of job applications, including application ID, user ID, job ID, and application status.

- Resume Table: Stores resume file details such as file name, file path, and associated user ID.
- OTP Table: Stores one-time passwords for email verification and password reset functionality.

Relationships are established using foreign keys, where the Application table references both User and Job tables. This structure ensures efficient data retrieval and supports features like application tracking, candidate shortlisting, and recruitment analytics.

The database schema is designed to support core functionalities such as user management, job posting, and application tracking. The main entities in the schema are User, Job, Application, Resume, and OTP. The Job entity contains details such as job title, description, required skills, salary, and location. The Application entity maintains the relationship between users and jobs.

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Chapter 4

METHODOLOGIES

4.1 Project Architecture

The Job Portal Management System follows a layered architecture where the user (job seeker or employer) first registers or logs into the system using valid credentials with email verification through OTP authentication. After successful login, users can perform actions such as browsing job listings, applying for jobs, or managing job postings.

These inputs are processed by the Spring Boot backend, where controllers and service layers validate the data and execute business logic. The system also integrates a chatbot module that assists users by answering queries and guiding them through various functionalities[8].

The application interacts with a MySQL database using JPA/Hibernate to store and retrieve user data, job details, application records, and resume information efficiently. Once processing is complete, the system generates a response, which is displayed to the user in the form

of job listings, application status updates, or confirmation messages. This ensures a smooth, secure, and efficient flow of data throughout the application.

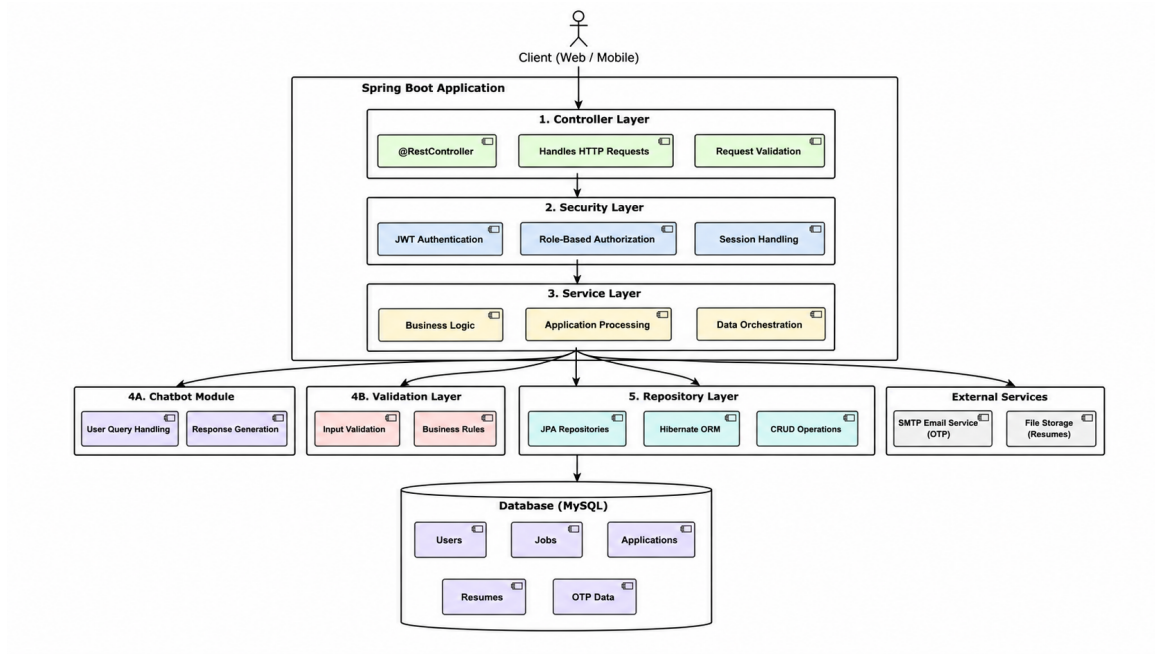


Figure 4.1: System Architecture Diagram

[Figure 4.1] represents the layered architecture of the Job Portal Management System built using Spring Boot. It shows how user requests flow from the frontend to the controller, security, service, and repository layers for processing. The system integrates modules like chatbot, validation, and email services to enhance functionality. Finally, all processed data is stored and retrieved from the MySQL database, ensuring efficient and secure operation.

4.2 DataFlow Diagram

The data flow of the Job Portal Management System begins when the user (job seeker or employer) registers or logs into the system using valid credentials along with email verification. After authentication, the user performs actions such as searching for jobs, applying for jobs, posting job listings, or managing applications.

These inputs are sent to the Spring Boot backend, where controllers and service layers validate the data and execute the required business logic. The chatbot module can also process user queries and provide real-time assistance.

The processed data is then stored in or retrieved from the MySQL database using JPA/Hibernate, ensuring efficient data management. Finally, the system generates a response, which is displayed to the user in the form of updated job listings, application status, or notifications, ensuring a continuous and efficient flow of data throughout the system. The system integrates modules like chatbot, validation, and email services to enhance functionality. Finally, all processed data is stored and retrieved from the MySQL database, ensuring efficient and secure operation.

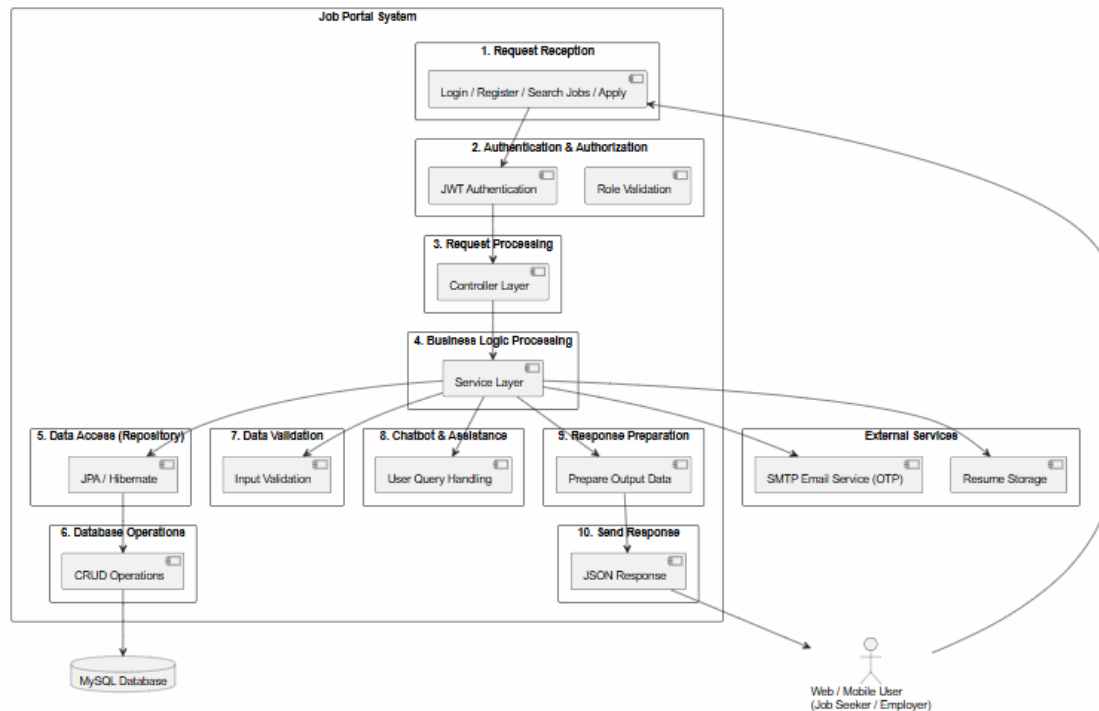


Figure 4.2: DataFlow Diagram

[Figure 4.2] illustrates the data flow in the Job Portal Management System, showing how user requests move through different processing stages. It begins with authentication and request handling, followed by business logic execution in the service layer. The system interacts with modules like validation, chatbot assistance, and external services such as email verification and resume storage. Finally, data is stored in the database and a response is sent back to the user.

4.3 Module 1: User Module

This module handles all functionalities related to job seekers. It allows users to register, log in using secure email verification (OTP), create and manage profiles, upload resumes, and browse available job listings.

Users can filter jobs based on criteria such as location, skills, or experience and apply for jobs online. The module also allows users to track their application status (applied, shortlisted, rejected). Additionally, an integrated chatbot assists users by providing instant support and guidance. The module ensures proper validation of user inputs and maintains records of user activities in the database.

4.4 Module 2: Employer Module

The employer module is responsible for managing job postings and candidate applications. Employers can securely log in and perform operations such as creating new job postings, updating job details, deleting job listings, and viewing applications submitted by job seekers.

Employers can shortlist or reject candidates based on their qualifications and manage recruitment processes efficiently. This module helps in monitoring candidate progress and improves hiring decision-making.

4.5 Module 3: Database Module

The database module manages data storage and retrieval. It includes entities such as User, Job, Application, Resume, and OTP. All CRUD operations are performed using Spring Data JPA, ensuring efficient interaction with the MySQL database.

This module maintains data integrity and supports features like job tracking, application management, resume storage, and secure email verification. It enables efficient querying for job filtering and candidate selection.

Advantages of this Project

This project provides a streamlined and digital approach to managing the recruitment process by connecting job seekers and employers on a unified platform.

a) User-Friendly Interface:

The system offers an intuitive and responsive interface, making it easy for users to browse jobs, apply online, and manage applications efficiently.

b) Secure Authentication:

The application uses SMTP-based email verification (OTP) and Spring Security to ensure secure login and data protection.

c) Real-Time Assistance:

An integrated chatbot provides instant support to users by answering queries and guiding them through the system.

d) Centralized Data Management:

All user, job, and application data are stored in a structured database, simplifying data access and management.

e) Efficient Recruitment Process:

Employers can manage job postings and applications effectively, improving hiring efficiency and decision-making.

Disadvantages

The system has certain limitations that can be addressed in future improvements. It currently depends on a local server setup, limiting accessibility beyond a specific environment. Advanced features such as AI-based job recommendations, mobile application support, and large-scale deployment are not yet implemented.

Additionally, performance optimization may be required to handle high user traffic and large volumes of job applications. Integration with external job APIs and advanced analytics can further enhance the system. The system integrates modules like chatbot, validation, and email services to enhance functionality. Finally, all processed data is stored and retrieved from the MySQL database, ensuring efficient and secure operation.

Chapter 5

RESULTS AND DISCUSSIONS

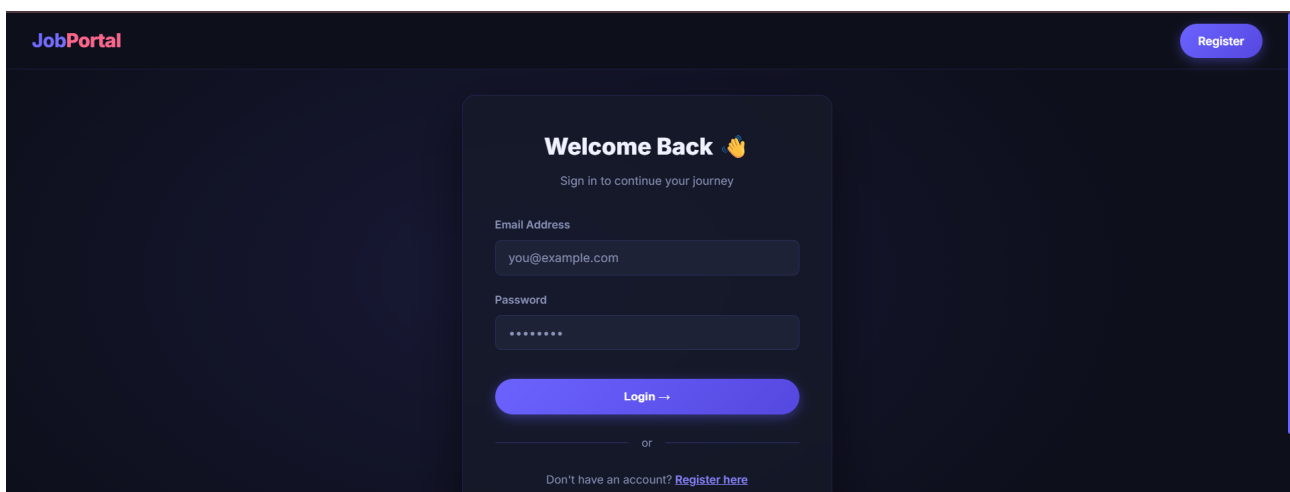


Figure 5.1: Login page for Employer and Student Authentication

[Figure 5.1] represents and shows the login page where students (job applicants) and employers (job providers) can enter their credentials to access the system. Users can also log in via Google authentication (Firebase), and based on their roles, they are redirected to their respective dashboards.

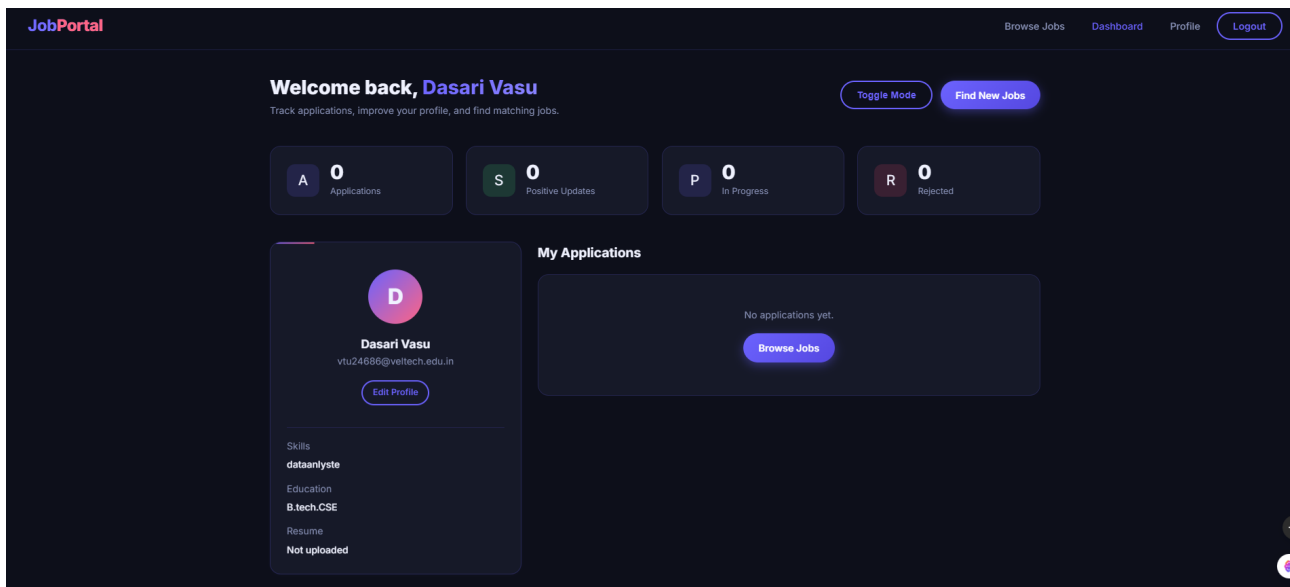


Figure 5.2: Student Dashboard

[Figure 5.2] represents the student dashboard (shown above) provides a centralized interface for managing all job-related activities efficiently. It enables users to track job applications in real time with clear status indicators such as applied, under review, shortlisted, and rejected, helping them monitor their progress easily. Students can upload and manage resumes, browse and apply for jobs based on their preferences, and update their profiles to improve visibility to employers. The dashboard also offers features like personalized job recommendations, notification alerts for application updates, and access to interview preparation tools. Additionally, an integrated chatbot assists users by providing instant guidance, answering queries, and offering suggestions to enhance their job search and application process. This interactive and user-friendly design ensures a smooth and productive

experience for job seekers.

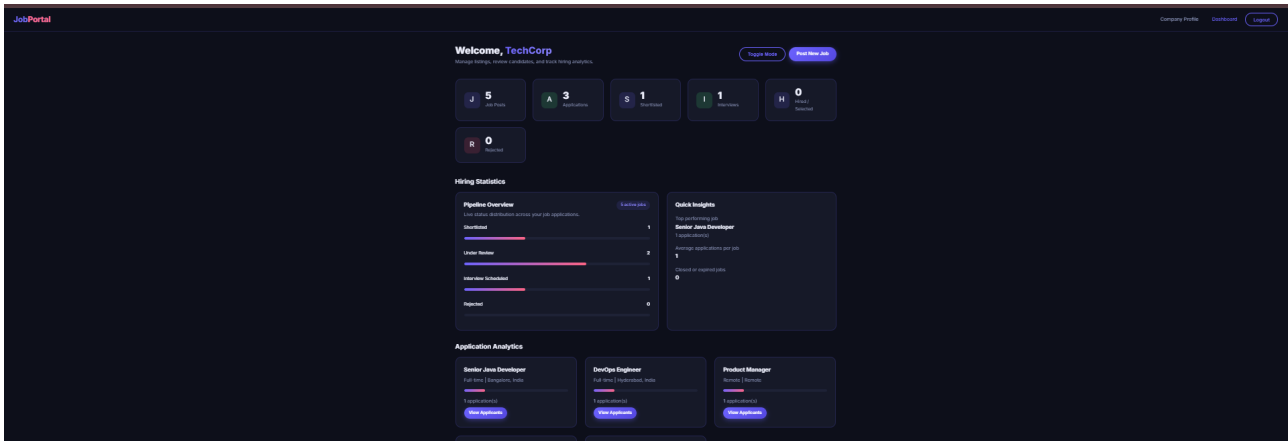


Figure 5.3: Admin Dashboard

[Figure 5.3] represents the admin (employer) dashboard (shown above), which provides a comprehensive overview of job postings and recruitment activities through an analytics-driven interface. It allows employers to create and manage job listings, monitor active and total job postings, and track the number of applications received. The dashboard presents key insights such as application trends over time, status distribution of candidates (under review, shortlisted, rejected), and performance metrics like average applications per job, response rate, and shortlist rate. Employers can also communicate with applicants, manage company profiles, and post new job openings directly from the interface. This centralized system enables efficient hiring decisions by offering real-time data visualization, streamlined candidate management, and improved recruitment workflow.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Job Portal Management System was successfully designed and implemented to simplify and digitize the recruitment process by connecting job seekers and employers on a unified platform. The system integrates a responsive frontend developed using HTML, CSS, and JavaScript with a Spring Boot backend, along with a MySQL database for efficient data storage and management. The application enables users to create profiles, upload resumes, search and apply for jobs, and track application status through an intuitive interface. The integration of a chatbot enhances user interaction by providing real-time assistance and guidance throughout the platform. Seamless communication between frontend and backend through REST APIs ensures efficient data processing and a smooth user experience. Compared

to existing large-scale recruitment platforms, the proposed system is lightweight, cost-effective, and tailored for academic or small-scale recruitment environments. It reduces manual effort, improves transparency, and provides a structured and efficient approach to managing job applications.

6.2 Future Enhancements

The current system provides a strong foundation but can be further enhanced to improve scalability, performance, and user experience. Future improvements include:

- Implementing AI-based job recommendation systems to provide personalized job suggestions to users
- Developing a mobile application to enhance accessibility across different devices
- Integrating advanced analytics dashboards for employers to track recruitment trends and performance
- Enhancing the chatbot with Natural Language Processing (NLP) for more intelligent and context-aware responses
- Implementing real-time notifications through email and SMS for application updates

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Table 7.1: Mapping of Project to Complex Engineering Problem (CEP)

Level	Attribute	Characteristics	Wks	Justification
WP1	Depth of Knowledge	Requires in-depth engineering knowledge	WK3, WK4, WK5, WK6	Requires knowledge of Spring Boot, Spring MVC, Spring Security, JPA, and database design for building a multi-role job portal system
WP2	Conflicting Requirements	Technical and non-technical conflicts	WK4, WK5, WK6	Balances role-based access control with usability, ensuring employers and job seekers have separate secure and intuitive workflows
WP3	Depth of Analysis	Analytical and design thinking	WK3, WK4, WK5	It involves entity-relationship modeling, REST API design, and integration of file upload and email notification services
WP4	Familiarity	Partially new problems	WK4, WK8	Job-matching and application tracking introduce modern challenges in multi-role web systems beyond traditional CRUD applications
WP5	Applicable Codes	Limited standard solutions	WK6	Requires custom design for resume storage, application status tracking, and dynamic job filtering not fully covered by standard frameworks
WP6	Stakeholder Needs	Multiple stakeholders involved	WK5, WK6, WK7	Handles distinct requirements of job seekers, employers, and admins, ensuring secure and efficient role-specific interactions
WP7	Interdependence	System-level integration	WK3, WK5, WK6	Integrates frontend (Thymeleaf), backend (Spring Boot), database (JPA/MySQL), file storage, security, and email notification into one unified system

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

Table 7.2: Mapping of Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Use of diverse technologies	The project uses Spring Boot, Spring MVC, Spring Data JPA, MySQL/H2, Thymeleaf, Spring Security, and Java-MailSender for full-stack portal development
EA2	Level of Interactions	Complex interactions between modules	The system manages layered interactions between job listings, applications, user authentication, resume uploads, and email notifications
EA3	Innovation	Application of modern technologies	Integration of role-based dashboards, dynamic job filtering with JavaScript/ES6, file upload handling via Multi-partFile, and email status notifications demonstrates innovation
EA4	Consequences to Society and Environment	Impact on users and society	Improves employment accessibility, reduces manual hiring overhead, and provides a transparent and secure digital platform for job seekers and employers
EA5	Familiarity	Beyond standard practices	Combines multi-role authentication, resume management, application tracking, and real-time dashboard analytics into a unified portal beyond traditional job boards

7.3 SDG Mapping (CEP Section)

Table 7.3: SDG Mapping for the Project

SDG No.	SDG Title	Relevance to the Project
SDG 8	Decent Work and Economic Growth	Facilitates employment by connecting job seekers with employers, supporting fair recruitment processes and improving career opportunities
SDG 9	Industry, Innovation and Infrastructure	Promotes digital transformation in recruitment using Spring Boot REST APIs, dynamic filtering, and automated notification systems
SDG 10	Reduced Inequalities	Provides a standardised, accessible digital platform for all job seekers regardless of geography, reducing barriers to employment opportunities
SDG 16	Peace, Justice and Strong Institutions	Ensures data security and role-based access control using Spring Security and password hashing, maintaining integrity and fairness in the hiring process

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